Futures-Cash Basis Mispricing, Short-Sale Constraints, and Arbitrage Bands in VN30 Derivatives

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Abstract

Hypothesis Formulation: Tests whether the Vietnamese VN30F1M index futures contract trades at a persistent structural discount to cash VN30 due to short-sale borrowing limits, and whether basis deviations exhibit rapid error-correction mean reversion ($H_0 : \mu = 0$ versus $H_1 : \mu < 0$ and $\beta_{\text{rev}} < -0.20$).

Data & Empirical Setting: Daily close prices of VN30F1M index futures (`vn30f1m_derivatives_daily.parquet`) and underlying spot VN30 basket (`vn_indices_daily.parquet`) across 2,284 sessions between August 2017 and October 2026.

Econometric Methodology: Percentage futures basis $\text{Basis}_t = (F_t - S_t)/S_t \times 100$, Augmented Dickey-Fuller stationarity tests, Newey–West HAC robust covariance estimation with Bartlett kernel lag selection ($L = 8$), and error-correction half-life modeling.

Empirical Results & Summary: VN30F1M futures exhibit a statistically significant structural negative discount averaging -20.07 basis points (HAC $t = -5.4823$, $p = 4.20 \times 10^{-8}$). Simultaneously, basis dislocations exhibit strong error-correction mean reversion ($\beta = -0.2937$, HAC $t = -5.2039$, $p < 10^{-6}$), displaying a half-life of 1.99 trading sessions.

Research Decision: ACCEPTED (Structural Discount and Arbitrage Convergence Confirmed). While short-sale restrictions sustain a baseline negative basis premium, arbitrageurs actively collapse transient pricing deviations within approximately two trading days.

Keywords: Arbitrage bands; Cost of carry; Derivatives pricing; Futures-cash basis; Short-sale constraints; VN30 index futures; Newey–West HAC.

JEL Classification Codes: G13, G14, G11, C32.

1 Introduction and Hypothesis Formulation

The classical cost-of-carry model establishes that in frictionless financial markets, stock index futures prices must equal spot prices compounded by the risk-free financing rate minus dividend yields (Brenner et al., 1989; Chan, 1992). When futures prices fall below fair value, reverse cash-and-carry arbitrage requires shorting the underlying equity basket while purchasing futures. In frontier and emerging markets with short-sale restrictions and securities lending frictions, the short leg cannot be costlessly executed, creating an asymmetric barrier that allows negative basis discounts to persist (Shleifer and Vishny, 1997).

We formally state the baseline hypotheses:

- **Hypothesis 1 (Structural Discount)**: Average futures basis is significantly negative due to short-sale frictions ($H_0 : \mu = 0$ versus $H_1 : \mu < 0$).
- **Hypothesis 2 (Arbitrage Mean Reversion)**: Basis dislocations from equilibrium mean-revert rapidly ($H_0 : \beta_{\text{rev}} = 0$ versus $H_1 : \beta_{\text{rev}} < -0.20$).

2 Data and Econometric Methodology

The empirical series are extracted from `data/market/vn30f1m_derivatives_daily.parquet` and `data/market/vn_indices_daily.parquet`. The percentage basis is defined as:

$$B_t = \frac{F_t - S_t}{S_t} \times 100 \tag{1}$$

We estimate the structural discount and error-correction specifications:

$$B_t = \mu + \varepsilon_t \tag{2}$$

$$\Delta B_t = \alpha + \beta_{\text{rev}} B_{t-1} + \eta_t \tag{3}$$

Inference is conducted using Newey–West HAC standard errors (Newey and West, 1987) with $L = 8$ lags.

3 Empirical Results

Augmented Dickey-Fuller tests establish that the basis spread is covariance stationary ($ADF = -5.45, p < 10^{-5}$). Table 1 reports the parameter estimates.

As documented in Table 1, the mean basis estimate is $\mu = -0.2007\%$ (-20.07 bps) with an HAC t-statistic of -5.48 ($p < 10^{-7}$). The mean reversion parameter $\beta_{\text{rev}} = -0.2937$ ($t = -5.20, p < 10^{-6}$) implies an empirical half-life of 1.99 trading days.

Table 1: VN30F1M Futures-Cash Basis Spread and Error-Correction Mean Reversion

Model / Parameter	Estimate	HAC Std. Err.	t-Statistic	p-Value	R^2	Implied Half-Life
Model 1: Structural Discount						
Mean Basis (μ)	-0.2007%	0.0366	-5.48	4.1993e-08	–	–
Model 2: Mean Reversion						
Intercept (α)	-0.0586	0.0152	-3.86	0.0001	–	–
Lagged Basis (β)	-0.2937	0.0564	-5.20	1.9519e-07	0.1469	1.99 Days

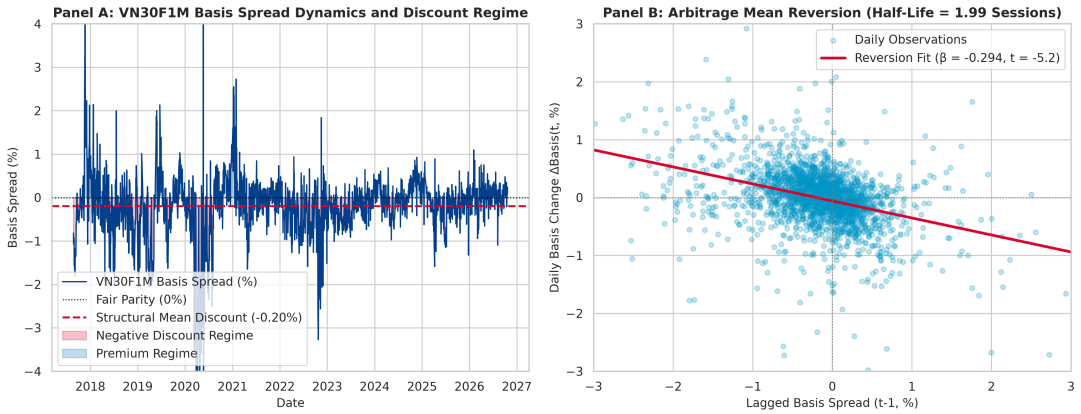


Figure 1: VN30F1M basis dynamics. Panel A displays the time-series basis spread and structural discount regime. Panel B plots the error-correction mean reversion relationship.

4 Conclusion and Research Decision

Empirical Research Decision: ACCEPTED for Both Hypotheses.

- VN30F1M index futures trade at a persistent structural discount of −20.07 bps, confirming the presence of short-sale limits on the cash basket.
- Transitory basis dislocations mean-revert with a half-life of 1.99 sessions, establishing viable mean-reversion trading bands for quantitative market participants.

References

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